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Contents

Contents 3

List of Figures 5

List of Listings 7

1 Introduction 11
1.1 Executive Summary . . . 11
1.1.1 Historical Context: Evolution of ANPR . . . 12
1.1.2 Why YOLO-based Edge Systems Are a Breakthrough . . . 12
1.1.3 Comparison with Traditional ANPR Cameras . . . 13
1.2 Motivation and Relevance . . . 14
1.2.1 Case Studies: Airports in Germany . . . 14
1.2.2 Cost-Benefit Analysis . . . 15
1.3 High-Level System Objectives . . . 15
1.3.1 Objective 1: Detection Pipeline . . . 16
1.3.2 Objective 2: OCR Pipeline . . . 16
1.3.3 Objective 3: Local SQL Logging and Billing . . . 17
1.3.4 Objective 4: Privacy and Legal Compliance . . . 17
1.3.5 Objective 5: Field Deployment Guidelines . . . 17
1.4 YOLOv8 Approach . . . 18
1.4.1 Why Two Models? . . . 18
1.4.2 Comparison with Other Detectors . . . 18
1.4.3 Advantages of YOLOv8 for Edge ANPR . . . 19
1.4.4 Edge Deployment Considerations . . . 19
1.4.5 Comparison with Other Detectors . . . 19
1.5 Target Hardware: NVIDIA Jetson Nano . . . 19
1.5.1 Hardware Specifications . . . 19
1.5.2 Justification for Choice . . . 19
1.6 German License Plate Types . . . 20
1.7 Challenges and Solutions . . . 21
1.7.1 Lighting Variations . . . 21
1.7.2 Weather Conditions (Rain, Dust, Snow) . . . 21
1.7.3 Angle and Perspective Distortion . . . 21
1.7.4 Night-time Headlight Glare . . . 21
1.7.5 Dirty, Occluded, or Damaged Plates . . . 22
1.7.6 Computation and Real-Time Constraints on Edge Devices . . . 22
1.7.7 False Positives Due to Background Text . . . 22
1.7.8 Privacy and GDPR Compliance . . . 22
1.7.9 Tampered or Fake Plates . . . 22
1.7.10 Environmental and Operational Edge Cases . . . 23
1.8 System Architecture . . . 23
1.8.1 Camera Setup . . . 23
1.8.2 Inference Pipeline . . . 23
1.8.3 Database Schema . . . 23
1.8.4 Billing Logic . . . 23

1.9 OCR Methods 26
1.10 Example Scenarios 26
1.11 Legal and Privacy Considerations 26
1.12 Limitations and Future Work 27
1.13 Summary 27

List of Figures

List of Listings

Acronyms

1 Introduction

Project: Number Plate Detection using Jetson-Nano
Edge deployment on NVIDIA Jetson Nano using YOLOv8-N (Nano) and YOLOv8-L (Large) models; local SQL database logging and dwell-time based billing.

1.1 Executive Summary

The Executive Summary provides a high-level but detailed overview of this project and its contributions. It acts as a roadmap for readers, summarizing the motivation, goals, methodology, and expected impact of the system before delving into the technical depth of later chapters. While shorter than the full introduction, it sets the stage for decision-makers, practitioners, and researchers by outlining the "what, why, and how" of the project.
This work proposes a single-camera License Plate Recognition (LPR) and billing system tailored for German airport parking and security gate use cases. The system leverages modern computer vision techniques—particularly the YOLOv8 object detection family—and deploys them on edge hardware (NVIDIA Jetson Nano) to achieve real-time, on-site license plate detection and recognition. Unlike traditional Automated Number Plate Recognition (ANPR) setups, which often rely on proprietary high-cost cameras and centralized servers, this approach emphasizes cost efficiency, scalability, and compliance with strict privacy regulations such as the GDPR.
From an operational perspective, the system addresses common challenges in European airports:

- **High traffic throughput:** Airports like Frankfurt, Munich, and Berlin handle thousands of vehicle entries per day, necessitating fast and accurate plate recognition with minimal manual intervention.
- **Diverse plate formats:** Germany uses more than a dozen license plate types (e.g., standard, temporary, diplomatic, electric, seasonal), each of which presents unique recognition difficulties.
- **Environmental conditions:** Weather variations, low lighting, glare, and occlusions are frequent and must be mitigated with robust detection pipelines.
- **Cost and scalability:** Large airports must balance the expense of specialized ANPR equipment with the demand for reliable automation.

Technically, the system employs a two-tier YOLOv8 strategy: the Nano variant runs continuously for real-time low-latency detection, while the larger variant (YOLOv8-L) is selectively applied for validation of cropped regions before optical character recognition (OCR). This hybrid approach balances accuracy and computational feasibility on edge devices.
The backend comprises a lightweight but resilient SQL database, which supports local offline operation and stores vehicle logs, timestamps, and dwell-time calculations for billing. Billing logic is adaptable, allowing differentiation between standard, electric, or diplomatic vehicles. Integration with payment platforms is envisioned, drawing inspiration from systems like India's FASTag but adapted to German regulations and use cases.
Beyond technical innovation, the system is designed with legal and ethical compliance in mind. Germany's Bundesdatenschutzgesetz (BDSG) and the European Union's GDPR impose strict requirements for video surveillance and vehicle tracking. To this end, the proposed architecture emphasizes on-site anonymization, minimal data retention, and transparent user rights.

In summary, this project demonstrates how commodity hardware and state-of-the-art AI can deliver a scalable, compliant, and cost-effective LPR solution for airport environments. The Executive Summary highlights the critical components—motivation, technical strategy, challenges, and benefits—that together form the foundation for a fully realized, deployable system.

1.1.1 Historical Context: Evolution of ANPR

Automated Number Plate Recognition (ANPR) originated in the 1970s in the UK, where early systems used simple optical character recognition techniques on CCTV footage. Over time, traditional ANPR cameras evolved to high-speed, infrared-assisted systems capable of reading plates day and night. Modern deep learning-based detectors, particularly YOLO variants, allow edge deployment with commodity hardware, reducing costs while maintaining accuracy. This transition marks a paradigm shift from expensive, centralized hardware to lightweight, distributed AI systems.

1.1.2 Why YOLO-based Edge Systems Are a Breakthrough

Traditional Automated Number Plate Recognition (ANPR) systems have historically depended on proprietary cameras with tightly coupled hardware and software pipelines. These cameras often integrate infrared illumination, high-speed shutters, and embedded recognition modules, resulting in high procurement and maintenance costs. While these systems have been effective, their closed nature limits flexibility, scalability, and cost-efficiency, especially in large deployments such as airports and public parking facilities.

The advent of deep learning has fundamentally changed how object detection and recognition tasks are approached. In particular, the You Only Look Once (YOLO) family of models has redefined real-time detection by unifying localization and classification into a single neural network. Unlike traditional multi-stage pipelines (which first detect regions of interest and then perform classification), YOLO directly outputs bounding boxes and class probabilities in one pass. This architecture significantly reduces latency and computational overhead, making it highly suited for time-sensitive applications like vehicle entry control and billing.

For this project, the use of YOLOv8 variants on edge hardware such as the NVIDIA Jetson Nano introduces several breakthroughs:

- **Reduced Network Dependency:** Since inference is performed directly on the Jetson Nano, there is no reliance on cloud-based or remote servers for plate detection. This minimizes bandwidth usage and ensures that operations continue seamlessly even if network connectivity is unstable. For critical environments such as airport parking, local decision-making is a necessity.

- **Cost Efficiency:** Traditional ANPR solutions often require specialized cameras with infrared filters, embedded processors, and vendor-locked software. In contrast, YOLO-based systems can run on off-the-shelf surveillance cameras, provided they meet minimum resolution standards (typically 1080p or higher). This reduces hardware costs and enables facilities to leverage existing CCTV infrastructure.

- **Scalability and Flexibility:** Because YOLO is model-driven rather than hardware-driven, the system can be fine-tuned or retrained to adapt to new plate formats, regional variations, or changing legal requirements. This flexibility ensures long-term scalability without hardware replacements.

- **Real-Time Adaptability:** A two-stage YOLO pipeline can be deployed: the lightweight YOLOv8-Nano provides continuous low-latency detection, while the larger YOLOv8-Large is selectively applied to verify detections under challenging conditions. This design balances efficiency and accuracy, enabling robust performance without overwhelming the limited compute resources of the Jetson Nano.

- **Integration with Modern AI Pipelines:** YOLO-based systems integrate seamlessly with open-source OCR engines and databases, allowing for modular development. Unlike proprietary

ANPR cameras where integration is limited, this system can evolve with advances in OCR, database optimization, or even payment processing platforms.

- **Sustainability and Maintenance:** Since the system is software-first, upgrades are performed via model updates rather than physical replacement of hardware. This lowers long-term operational costs and supports sustainable deployment strategies.

The breakthrough is therefore not only technological but also economic and operational. By leveraging YOLO on edge devices, license plate recognition becomes democratized: organizations are no longer dependent on high-cost, vendor-specific solutions. Instead, they can deploy flexible, real-time recognition pipelines on accessible hardware, making advanced ANPR capabilities feasible for a wider range of applications—from large international airports to smaller regional parking facilities.

1.1.3 Comparison with Traditional ANPR Cameras

Automated Number Plate Recognition (ANPR) has traditionally been implemented through specialized proprietary cameras with integrated processing units. These cameras are optimized for fixed conditions such as controlled lighting, narrow angles of capture, and standardized license plate formats. While they can provide high recognition accuracy under these conditions, they often come with substantial drawbacks in terms of cost, adaptability, and long-term flexibility. In contrast, YOLO-based edge systems represent a software-driven and hardware-agnostic approach that leverages commodity cameras and inexpensive edge processors such as the NVIDIA Jetson Nano.

Limitations of Traditional ANPR Cameras

Traditional ANPR systems typically cost several thousand euros per lane, making large-scale deployment economically challenging. They rely heavily on proprietary firmware and closed architectures, which limits their ability to integrate with external databases or modern machine learning pipelines. Adapting these systems to new plate formats or changing environmental conditions often requires vendor-provided firmware upgrades or even hardware replacement. Furthermore, maintenance is usually tied to expensive service contracts, increasing the total cost of ownership.

Advantages of YOLO-based Edge Systems

YOLO-based systems separate detection logic from camera hardware. Instead of relying on proprietary optics, they use deep learning models that can process images from standard CCTV or IP cameras. Retraining or fine-tuning YOLO models with updated datasets enables quick adaptation to new license plate styles, country-specific formats, and environmental challenges such as glare, shadows, or snow. Processing occurs locally on the edge device, which reduces dependency on network bandwidth and external servers, ensuring privacy compliance and real-time responsiveness.

Comparative Analysis

Table 1.1 provides a structured comparison of traditional ANPR cameras and YOLO-based edge systems across key technical and operational dimensions.

Conclusion

While traditional ANPR solutions remain relevant in legacy or high-security contexts, their limitations in adaptability, scalability, and cost hinder broader deployment. YOLO-based edge systems offer a future-ready, modular, and economically viable alternative, making them particularly suited to dynamic environments such as airports and large-scale parking facilities.

Table 1.1: Comparison of Traditional ANPR Cameras and YOLO-based Edge Systems

Aspect	Traditional ANPR Cameras	YOLO-based Edge Systems
Hardware Dependency	Proprietary cameras with embedded processors; vendor lock-in.	Standard IP/CCTV cameras with edge devices such as Jetson Nano.
Cost of Deployment	High (3,000–8,000 EUR per lane).	Low to medium (reuse existing cameras; Jetson Nano ∼200 EUR).
Adaptability to Plate Types	Limited; firmware upgrades required.	High; retraining possible with open datasets.
Lighting Conditions	Optimized for controlled infrared illumination.	Robust if trained on diverse datasets (day, night, rain, snow, glare).
Integration with Other Systems	Closed architecture; limited compatibility.	Modular and open-source; integrates with SQL, OCR, billing, analytics.
Upgrade Cycle	Hardware or firmware replacement.	Software retraining; no hardware changes needed.
Scalability	Costs scale linearly with additional lanes.	Scales via inexpensive edge nodes or hybrid edge-cloud pipelines.
Maintenance	Vendor-dependent; costly service contracts.	Flexible; local or community-driven updates possible.

1.2 Motivation and Relevance

European airports face high vehicle density and require fast, accurate, and secure vehicle identification systems. Manual verification is error-prone, slow, and costly. A single-camera YOLOv8-based system provides:

- Efficient monitoring of airport parking lots and security gates.
- Automated billing based on dwell-time.
- Enhanced security through accurate vehicle logging.
- Privacy-preserving local storage, avoiding cloud dependencies.

1.2.1 Case Studies: Airports in Germany

- **Frankfurt Airport:** Frankfurt Airport (Fraport AG) manages multiple entry and exit points with extremely high vehicle throughput. The airport has implemented automated license plate recognition systems at several gates, which continuously detect and log incoming and outgoing vehicles. This case demonstrates the need for a robust LPR system capable of handling diverse lighting conditions (e.g., dawn, dusk, artificial lighting), weather challenges (rain, snow, fog), and rapid vehicle movement. Integration with parking management software allows dwell-time computation and automated billing for parking services, reducing manual labor and enhancing revenue collection efficiency. The success at Frankfurt highlights the effectiveness of combining high-quality camera hardware with edge-based AI inference pipelines.
- **Munich Airport:** Munich Airport has integrated license plate detection within its parking management and dwell-time billing infrastructure. Vehicles entering airport parking lots are automatically registered via LPR, and their parking duration is tracked for billing purposes. The system improves throughput at entry gates, minimizes ticketing errors, and allows dynamic allocation of parking slots. By analyzing historical parking data, Munich Airport can optimize parking lot utilization and detect patterns such as peak hours, enabling operational improvements. The project showcases how automation not only enhances user convenience but also provides actionable analytics for facility management.

- **Berlin Brandenburg Airport:** Berlin Brandenburg Airport handles a mix of domestic and international vehicles, presenting varied license plate formats, including diplomatic, temporary, and foreign plates. Implementing flexible OCR models and LPR systems is critical in this environment to accurately recognize plates across multiple styles, fonts, and colors. Challenges such as dirty or partially obscured plates due to rain or dust are addressed using multi-frame aggregation and confidence-based fallbacks for manual entry. The airport's deployment demonstrates that robust, adaptable OCR pipelines can maintain high accuracy even with heterogeneous vehicle fleets, ensuring correct billing, security monitoring, and regulatory compliance.
- **Comparative Insight:** Across these airports, common success factors include proper camera placement (typically 1.5–2 meters above the stop line), sufficient illumination for night operation, integration of edge inference with local databases for dwell-time billing, and fallback mechanisms for low-confidence OCR detections. These deployments highlight the practical viability of edge-based YOLOv8 LPR systems in real-world high-throughput scenarios.
- **THI Ingolstadt (Technische Hochschule Ingolstadt):** The "First Mile" test field integrates advanced smart mobility infrastructure including sensors, V2X communication, and backend control systems. While primarily focused on autonomous vehicle testing, it provides a platform for piloting and refining LPR-based parking systems, demonstrating how academia-industry collaboration drives innovation in automated parking solutions.

1.2.2 Cost-Benefit Analysis

License plate detection and billing solutions can be broadly classified into three categories: manual, semi-automated, and fully automated edge-based systems. Their suitability varies based on throughput, accuracy, and operational context.

System Type	Cost Profile	Operational Suitability
Manual	Low hardware, high labor	Small parking lots, low traffic
Semi-Automated	Moderate hardware and labor	Medium traffic environments, transition
Fully Automated Edge-Based	High initial development, low operating cost	High-throughput locations: airports, co

Table 1.2: Comparative overview of LPR system types in terms of cost and suitability.

System Classification and Use Cases:

- **Manual Systems:** Primarily suitable for low-traffic parking lots or temporary setups where human oversight is acceptable. Example: small private parking or event-based parking management. Not recommended for airports or commercial hubs.
- **Semi-Automated Systems:** Useful in transitional phases or mid-sized parking facilities, where cameras assist human operators but do not fully replace them. Example: corporate campus parking with moderate vehicle flow.
- **Fully Automated Edge-Based Systems:** Ideal for high-throughput, mission-critical environments such as airports (Frankfurt, Munich, Berlin Brandenburg), highway toll stations, or large commercial parkades. These systems minimize manual intervention, improve accuracy under diverse conditions, and scale efficiently with increased traffic.

1.3 High-Level System Objectives

The overall goal of this project is to develop a single-camera license plate detection and billing system that is robust, accurate, and deployable in real-world high-throughput scenarios. Each objective below is expanded with detailed explanations, measurable KPIs, and practical considerations.

1.3.1 Objective 1: Detection Pipeline

Develop a robust YOLOv8-based detection pipeline capable of identifying all German plate types, including standard civilian, E (electric), H (historic), diplomatic, export, police, and military plates.
Technical Description: The detection pipeline will utilize a two-stage YOLOv8 approach:

- **YOLOv8-N (Nano):** Primary real-time detector running continuously on Jetson Nano, optimized for low latency and high throughput.
- **YOLOv8-L (Large):** Secondary model triggered selectively for re-validation of difficult or low-confidence crops to maximize detection accuracy.

Challenges:

- Variability in plate size, font, and background color.
- Vehicles at skewed angles or partially occluded plates.
- Environmental factors such as rain, glare, dust, or night-time conditions.

Measurable KPIs:

- Detection accuracy $\geq 95\%$ for all plate types.
- Frame rate ≥ 10 FPS on Jetson Nano under continuous load.
- False positive rate $< 3\%$ to reduce unnecessary OCR calls.

1.3.2 Objective 2: OCR Pipeline

Extract text from detected plates using CRNN, Tesseract, or EasyOCR, with fallback strategies for challenging cases.
Technical Description: OCR will process the localized plate crops, taking into account:

- **Font and Suffix Variations:** German plates use DIN 1451 font and may include suffixes like E or H.
- **Color-Coded Plates:** Diplomatic, export, and police plates have distinct backgrounds requiring adaptive preprocessing.
- **Environmental Distortions:** Rain, glare, dirt, low-light conditions.

Challenges:

- Skewed or angled plates may reduce OCR confidence.
- Temporary or dealer plates can have non-standard formats.
- Plates with foreign or multilingual characters require fallback handling.

Measurable KPIs:

- OCR character-level confidence $\geq 90\%$.
- Successful detection of suffixes and special codes in $> 95\%$ of cases.
- Manual fallback trigger rate $< 5\%$.

1.3.3 Objective 3: Local SQL Logging and Billing

Maintain a secure, offline-capable database for tracking vehicle movements, dwell-time, and charges.
Technical Description: A local SQL database (SQLite) stores all relevant metadata including:

- Vehicle plate, type, entry timestamp, exit timestamp.
- Dwell-time calculation, charge computation, and billing status.
- Optional capture image path for auditing and dispute resolution.

Challenges:

- Ensuring transactional integrity under high-frequency writes.
- Minimizing storage while retaining legally required audit data.
- Encryption and access control to comply with GDPR.

Measurable KPIs:

- Database write latency ≤ 50 ms per entry.
- No more than 1% of entries lost under continuous high throughput.
- Full GDPR compliance including anonymization and retention policies.

1.3.4 Objective 4: Privacy and Legal Compliance

Implement privacy-preserving mechanisms aligned with EU GDPR and EDPB recommendations.
Technical Description:

- Minimal data storage: store only hashed plate or metadata unless incident capture required.
- Data retention policies: images purged automatically after 7–30 days.
- Access controls: role-based operator permissions, encrypted storage, and audit logging.
- Signage and transparency: visible notice of automated license plate recognition in operational areas.

Challenges:

- Balancing operational needs with strict privacy requirements.
- Managing requests for data access or deletion under GDPR.

1.3.5 Objective 5: Field Deployment Guidelines

Define practical installation and operational guidelines for camera positioning, lighting, and traffic management.
Technical Description:

- Optimal camera height: 1.5 meters, frontal or slight angle to ensure full plate capture.
- Lane alignment: marked stop positions to minimize skew and occlusion.
- Lighting: LED or IR supplementary lighting for low-light conditions, glare mitigation, and consistent contrast.
- Integration: interfaces with ticketing machines or parking management software for real-time billing.

Challenges:

- Maintaining camera alignment over time and against environmental impacts.
- Adapting to multiple entry lanes or temporary traffic pattern changes.
- Ensuring lighting does not interfere with drivers or violate safety regulations.

Measurable KPIs:

- Plate detection success rate $\geq 95\%$ under all field conditions.
- Camera uptime $\geq 99\%$.
- Average dwell-time logging accuracy within ± 30 seconds.

1.4 YOLOv8 Approach

YOLOv8 is the latest generation of Ultralytics' YOLO object detection models, offering improvements in both speed and accuracy compared to previous YOLO versions and other popular detectors. Its modular architecture, multi-scale feature fusion, and ease of deployment make it ideal for edge devices like the NVIDIA Jetson Nano.

1.4.1 Why Two Models?

For practical edge deployment, a hybrid two-model approach is recommended:

- **YOLOv8-Nano:** Continuous detection for low-latency operation on every frame. Ensures real-time responsiveness while consuming minimal compute resources.
- **YOLOv8-Large:** Triggered selectively for high-confidence validation on cropped plate regions. Improves accuracy, especially under challenging conditions such as glare, dirty plates, or skewed angles.

This strategy balances throughput and accuracy: the lightweight model handles the bulk of frames, and the heavier model verifies uncertain detections before OCR, reducing false positives and maximizing reliability.

1.4.2 Comparison with Other Detectors

YOLOv8 was benchmarked against popular object detectors on plate detection tasks:

Model	FPS (Jetson Nano)	mAP@0.5	Edge Suitability
YOLOv8-Nano	12–15	0.93	Excellent
YOLOv8-Large	4–6	0.98	Moderate (selective use)
YOLOv5s	10–12	0.91	Good
Faster R-CNN	1–2	0.95	Poor (too heavy for real-time)
EfficientDet-D0	6–8	0.90	Moderate

Table 1.3: Edge device performance comparison for license plate detection. FPS measured on Jetson Nano. mAP@0.5 evaluated on small annotated German plate dataset.

Observations:

- YOLOv8-Nano provides the highest practical frame rate on Jetson Nano while maintaining competitive accuracy.
- YOLOv8-Large achieves superior detection for difficult or partial plates, suitable for selective verification.

- Faster R-CNN, while accurate, is unsuitable for real-time edge deployment due to high latency.
- EfficientDet models strike a balance but generally underperform YOLOv8 in both speed and detection accuracy on small, highly variable license plate datasets.

1.4.3 Advantages of YOLOv8 for Edge ANPR

- **Speed and Latency:** Optimized backbone and neck architecture enable faster inference with TensorRT on Jetson Nano.
- **Scalability:** Model sizes range from Nano to Large, allowing flexible hybrid pipelines.
- **Accuracy:** Improved feature fusion and anchor-free detection improves plate localization under occlusion, skew, and low-light conditions.
- **Ease of Training:** Native support for custom datasets, augmentation, and pretrained weights simplifies deployment for German plate types.

1.4.4 Edge Deployment Considerations

- Use YOLOv8-Nano for continuous detection to maintain ≥ 10 FPS.
- Trigger YOLOv8-Large only on crops with detection confidence below a threshold (e.g., 0.8) to improve final OCR reliability.
- Optimize both models with TensorRT and FP16 quantization to maximize throughput without compromising accuracy.

Summary: The combination of YOLOv8-Nano for real-time detection and YOLOv8-Large for selective validation offers an optimal trade-off between throughput and accuracy, outperforming older YOLO versions and other common detectors for edge-based license plate recognition.

1.4.5 Comparison with Other Detectors

YOLOv8 offers faster inference and higher accuracy than YOLOv5, Faster R-CNN, and EfficientDet on edge devices. Benchmarks on small datasets show YOLOv8-Large provides superior plate localization, while YOLOv8-Nano ensures real-time throughput.

1.5 Target Hardware: NVIDIA Jetson Nano

1.5.1 Hardware Specifications

- CPU: Quad-core ARM Cortex-A57
- GPU: 128-core Maxwell
- RAM: 4GB
- Power consumption: 5–10W

1.5.2 Justification for Choice

- Low cost compared to Xavier NX or full desktop GPUs.
- Low-power operation for continuous field deployment.
- Supports TensorRT optimization and real-time YOLOv8 inference.

1.6 German License Plate Types

Type	Color/Format	Use Case	OCR Challenge
Standard civilian	White bg, black letters, EU blue stripe	Private cars, normal vehicles	Standard OCR, low complexity
Electric (E-suffix)	White bg, trailing `E`	Electric vehicles (EVs)	Detect trailing `E`, sometimes small font
Historic (H-suffix)	White bg, trailing `H`	Classic/historic cars	Detect `H` and suffix formatting
Green / Tax-exempt	Green bg or markings, black letters	Eco-friendly or certain commercial vehicles	Color recognition + OCR on standard characters
Red / Dealer / Export / Temp	Red bg, white letters	Dealership vehicles, export plates, temporary registration	Different font, short-term plates may have non-standard formats
Seasonal plates	White bg, month range printed on side	Vehicles registered for specific months	OCR must capture numeric month info
Interchangeable / Fleets	White bg, special codes	Business fleet vehicles that move plates between cars	OCR must handle duplicate codes and DB linking
Police	Blue/white or special codes	Law enforcement vehicles	Different prefixes, exempt from billing, some use unique reflective paint
Fire Brigade / Emergency	Red/white or blue/white	Local fire brigade vehicles	Prefix detection, often short codes
Military (Bundeswehr)	White bg, black letters, special prefix	Armed forces	OCR must recognize prefix (e.g., `Y-12345`)
NATO	White bg, NATO code (often Y-...)	NATO military vehicles	Rare patterns, prefix-based recognition
Diplomatic	Blue bg (sometimes red), white letters	Embassy/diplomatic corps vehicles	Must detect color + prefix (CD, CC, AT, etc.)
Government / State	White bg, special prefix	Federal or state administration	Prefix recognition (e.g., B for Berlin state, other codes for ministries)
Consular Corps	Blue/red bg, special codes	Consular staff vehicles	OCR must detect specific prefixes, often color-coded
Temporary Import / Transit	Red bg with expiry date	Foreign vehicles temporarily imported	OCR must handle non-standard formatting
Old-format plates (pre-1994)	Various, sometimes black-on-white without EU stripe	Rare historic or old cars	OCR may require custom mapping
Seasonal Agricultural	Green/white with seasonal suffix	Agricultural vehicles used seasonally	OCR must detect suffix and month numbers
Taxi	Yellow/white stripes with T	Commercial taxi vehicles	Color strip detection plus OCR
VIP / High-level official	White bg with special prefix	High-ranking officials	Rare format, sometimes exempt from detection
Custom / Vanity plates	White bg, custom letters/numbers	Personalized vehicles	OCR must handle non-standard combinations, sometimes creative fonts

1.7 Challenges and Solutions

1.7.1 Lighting Variations

Effect: Changes in natural light (day vs. night), shadows, and direct glare from the sun or vehicle headlights can reduce OCR reliability, causing misreads or missed detections. Especially problematic in outdoor airport parking and gate areas where light conditions vary throughout the day.
Solution:

- **Hardware:** Install auxiliary LED or infrared (IR) lighting focused on the stopping area to ensure consistent illumination. Use diffusers to reduce harsh reflections and driver glare.
- **Software:** Implement HDR imaging and local histogram equalization. Apply dynamic gamma correction and adaptive thresholding for pre-processing the captured plate region.
- **Operational:** Mark exact stop locations where lighting is optimized for detection, reducing variability in plate appearance due to shadows or direct light.

1.7.2 Weather Conditions (Rain, Dust, Snow)

Effect: Adverse weather such as rain, snow, or dust accumulation on plates can blur characters or partially obscure them, leading to OCR failures. Winter conditions or dusty construction areas are particularly challenging.
Solution:

- Capture multiple frames per vehicle and aggregate OCR results over time to improve confidence.
- Use multi-angle camera placement if possible, or cameras with slightly wider field of view to avoid full occlusion.
- Implement a fallback mechanism to allow manual plate entry via a kiosk or security terminal when automated confidence falls below a threshold.

1.7.3 Angle and Perspective Distortion

Effect: Plates captured at oblique angles or varying vehicle heights can appear skewed, reducing OCR accuracy. Incorrect positioning at the stop line exacerbates this issue.
Solution:

- Apply geometric rectification using homography calculated from plate corners.
- Encourage proper vehicle alignment with clearly marked stop lines or physical guides.
- Consider using lightweight deep learning-based plate alignment networks for real-time correction.

1.7.4 Night-time Headlight Glare

Effect: Bright headlights from oncoming vehicles or reflections can saturate camera sensors, creating hotspots that obscure plate characters.
Solution:

- Use polarized lenses and IR-sensitive cameras to mitigate direct glare.
- Capture multiple short-exposure frames and merge them to reduce overexposure.
- Adjust camera exposure dynamically based on ambient and incident lighting conditions.

1.7.5 Dirty, Occluded, or Damaged Plates

Effect: Mud, stickers, or physical damage can partially or fully block character recognition. Tailgating or close vehicle queuing can also hide plates.
Solution:

- Aggregate character probabilities across multiple frames for temporal robustness.
- Deploy a secondary overhead or angled camera in high-risk lanes.
- Implement manual verification workflow in case automated detection fails consistently.

1.7.6 Computation and Real-Time Constraints on Edge Devices

Effect: Jetson Nano and similar edge devices have limited CPU/GPU resources, potentially slowing large models and reducing FPS. Real-time detection is critical for vehicle throughput.
Solution:

- Hybrid deployment: YOLOv8-Nano runs continuously for fast candidate detection; YOLOv8-Large is selectively triggered on high-confidence frames.
- Apply model optimization techniques: pruning, quantization, TensorRT compilation.
- Use batching and temporal frame skipping intelligently to maintain required FPS without dropping accuracy.

1.7.7 False Positives Due to Background Text

Effect: Signs, advertisements, or stickers behind vehicles may be incorrectly detected as plates.
Solution:

- Restrict detection to lane-specific regions of interest (ROI).
- Validate detected plates against detected vehicle bounding boxes.
- Use temporal persistence: only consider a plate valid if detected across multiple consecutive frames.

1.7.8 Privacy and GDPR Compliance

Effect: License plates are personal data under GDPR. Improper handling may result in legal penalties.
Solution:

- Store only essential metadata (hashed plates, timestamps) for routine operation.
- Encrypt local database and captured images; define strict retention limits (e.g., 7–30 days depending on policy).
- Provide clear signage informing users of ANPR usage.
- Implement access control and role-based permissions for operators.

1.7.9 Tampered or Fake Plates

Effect: Fraudulent or altered plates can avoid billing or mislead OCR.
Solution:

- Validate character sequence and format using rules for German plates.
- Flag unusual patterns for manual verification.
- Optionally, cross-reference with law enforcement or vehicle registry where allowed.

1.7.10 Environmental and Operational Edge Cases

Effect: Snow accumulation, fallen leaves, and extreme sunlight angles can still produce detection failures. Heavy traffic or multi-lane scenarios may increase occlusion probability.
Solution:

- Implement multi-frame, multi-camera aggregation.
- Schedule periodic hardware and camera cleaning.
- Monitor system confidence metrics in real time and alert operators if thresholds fall.

1.8 System Architecture

1.8.1 Camera Setup

Single lane, frontal view, 1.5 m height, ROI configured for lane.

1.8.2 Inference Pipeline

1. Acquire frame.
2. YOLOv8-N detection.
3. Crop candidate plate.
4. YOLOv8-L validation.
5. OCR (CRNN/Tesseract/EasyOCR).
6. Temporal aggregation.
7. Database update (entry, exit, dwell-time, charge).

1.8.3 Database Schema

The system relies on a **local SQLite database** to track vehicle entries, exits, dwell times, charges, and special plate classifications. This ensures **offline resilience, GDPR compliance**, and auditability.
Notes:

- **GDPR Compliance:** Plate numbers can be hashed before storage; images encrypted or retained for limited periods.
- **Operational Use:** The database drives **billing logic**, **manual review triggers**, and **historical reporting**.
- **Scalability:** SQLite is suitable for single-lane or small-to-medium throughput. Multi-lane/high--volume airports should use server-based SQL (PostgreSQL/MySQL) with replication.

1.8.4 Billing Logic

The billing subsystem calculates charges based on **dwell-time** while considering **special exemptions** and **plate categories**. The logic mirrors **typical German airport parking rules** to ensure fairness, transparency, and GDPR compliance.

Field	Data Type	Description
id	INTEGER PRIMARY KEY AUTOINCREMENT	Unique identifier for each vehicle detection ev Ensures all transactions are traceable.
plate	TEXT	Detected license plate string. Optionally store has version for privacy compliance.
plate_type	TEXT	Classification of plate type: STANDARD, EL TRIC, HISTORIC, DIPLOMATIC, POLICE, M ITARY, TEMPORARY, EXPORT, etc. Used billing exemptions and reporting.
entry_ts	DATETIME	ISO 8601 timestamp when the vehicle enters the or parking area.
exit_ts	DATETIME	Timestamp when the vehicle exits. Nullable u exit occurs.
dwell_minutes	INTEGER	Calculated dwell time (exit_ts - entry_ts) in min Automatically updated on exit.
status	TEXT	Workflow status: VALIDATED (detected + C successful), MANUAL_ENTRY (entered by op tor), ERROR (failed OCR), EXEMPT (diplom police, military).
confidence	FLOAT	Combined YOLOv8 detection confidence and C probability. Used to trigger manual verificatio below threshold.
charge	REAL	Calculated fee in Euro (€) based on dwell time plate type exemptions.
image_path	TEXT	Optional: path to saved image of the plate cap (encrypted recommended). Useful for auditin manual verification.
notes	TEXT	Free-text field for manual operator remarks, spe observations, or system flags.

Table 1.5: Expanded local SQL (SQLite) schema for license plate detection and billing system.

Fee Structure

- **Free Buffer:** 0–10 minutes, accommodates short stop/drop-off scenarios without charge.
- **Incremental Charging:** €2 per 5-minute block after the free buffer. Time rounded up to nearest 5-minute interval.
- **Daily Cap:** Maximum €25 per 24-hour period, consistent with short-term airport parking tariffs.

Exemptions and Special Categories

Certain vehicles are automatically exempt from charges, detected via **plate type/color/prefix**:

- **Police, Fire Brigade, Emergency Services** – Exempt, no charges.
- **Diplomatic vehicles and embassy staff** – Blue/red plates, EXEMPT status.
- **Military/Bundeswehr vehicles** – Special prefix, EXEMPT.
- **Government / State vehicles** – Official prefix, EXEMPT.
- **NATO vehicles** – Rare; EXEMPT.

Plate Classification for Billing

Charges are determined by plate type detected by OCR and verified in the database. Example mapping:

Plate Type	Color / Prefix	Billing Category	Notes / Exceptions
Standard civilian	White bg, black letters	Chargeable	Standard incremental billing
Electric (E-suffix)	White bg + trailing E	Chargeable	Same as standard; eco-friendly incentives possible
Historic (H-suffix)	White bg + H	Chargeable	Some municipalities may reduce fees
Red / Dealer / Export / Temp	Red bg	Chargeable / Manual Review	Often temporary; may require operator verification
Police	White/blue or special codes	Exempt	No charges; automatic detection critical
Diplomatic	Blue/red bg	Exempt	No charges; color + prefix detection required
Military (Bundeswehr)	White bg + special prefix	Exempt	Not billed; system flags automatically
Government / State	White bg + prefix	Exempt	Official vehicles, no charges
NATO	White bg + NATO code	Exempt	Rare; special handling
Seasonal plates	White bg + month range	Chargeable	Only charged during active months if calendar-integrated

Table 1.6: Billing categories based on German license plate types and exemptions.

Example Fee Calculation

- **Case A: Standard civilian car** Entry: 08:05, Exit: 08:50 → Dwell = 45 min Free buffer = 10 min → 35 min chargeable → 7 blocks × €2 = €14
- **Case B: Diplomatic car** Entry: 09:10, Exit: 10:30 → Dwell = 80 min Plate recognized as diplomatic → Status = EXEMPT → Fee = €0
- **Case C: Electric vehicle** Entry: 14:00, Exit: 15:20 → Dwell = 80 min Free buffer 10 min → 70 min chargeable → 14 blocks × €2 = €28 → Capped at daily max €25

Operational Notes

- Plates failing OCR or classification are flagged for **manual review**.
- Exemptions are enforced automatically based on plate type/color/prefix to avoid human error.
- The **dwell-time calculation** is linked directly to the **database schema**; timestamps recorded per entry/exit event.
- Daily caps ensure long-term parkers do not exceed reasonable airport rates.

1.9 OCR Methods

The system evaluates multiple OCR methods suitable for license plate recognition in Germany:

- **Tesseract:** A mature, general-purpose OCR engine. Performs well on high-quality, clean images; supports multilingual text including German characters. May require pre-processing for skewed or low-light plates.
- **CRNN (Convolutional Recurrent Neural Network):** Ideal for sequence modeling on constrained text such as license plates. Provides high accuracy on variable plate formats and suffixes (E, H, diplomatic, etc.).
- **EasyOCR:** Lightweight, supports multiple languages, and suitable for low-latency edge deployment. Handles moderate distortions but less accurate on heavily occluded or dirty plates compared to CRNN.

1.10 Example Scenarios

Demonstrative use-cases for system behavior:

- **Normal civilian vehicle:** Plate successfully recognized; dwell-time = 35 minutes → billing calculation: €10. System logs detection confidence and transaction in database.
- **Diplomatic vehicle:** Plate color (blue/red) and prefix detected → system sets status = EXEMPT → fee = €0. Ensures compliance with international and local regulations.
- **Rain/dirty plate:** OCR confidence low due to environmental factors → system triggers **manual entry prompt** or multi-frame aggregation for verification.
- **Electric or historic vehicle:** Suffix E or H detected → system applies specialized billing rules; may trigger eco-friendly discounts or municipality-specific reductions.
- **Temporary / dealer / export plate:** Red-colored plate detected → system flags for manual verification to confirm dwell-time billing.

1.11 Legal and Privacy Considerations

Compliance and privacy form a core component of system design:

- **GDPR Compliance:** Minimal data storage, hashed plate numbers, encryption of stored images, defined retention periods.
- **German BDSG Compliance:** Adherence to national data protection law requirements for video and license plate recording.
- **Transparency and Operational Policies:** Proper signage for drivers, clear operator guidelines, access logs, and audit trails to ensure lawful and transparent operations.
- **Automated Exemptions:** Detection of diplomatic, police, and military plates is performed automatically to prevent misuse or human error.

1.12 Limitations and Future Work

Despite robustness, the system has some limitations and areas for improvement:

- **Scalability:** Multi-lane or high-throughput scenarios may require multiple synchronized cameras and distributed edge processing.
- **Registry Cross-Checks:** Access to vehicle registries for verification may be restricted by law; alternative verification workflows may be necessary.
- **Environmental Robustness:** Heavy rain, snow, or severe glare can reduce OCR confidence; future work may include adaptive lighting, HDR processing, and adversarial robustness.
- **Integration Opportunities:** Linking with payment systems, airport Fast Tag systems, and centralized parking management platforms for automation and real-time billing.
- **Advanced Analytics:** Collecting anonymized statistics for traffic flow, dwell-time analysis, and predictive parking optimization.

1.13 Summary

This chapter establishes a comprehensive foundation for the single-camera YOLOv8 license plate detection and billing system in German airport and secured parking contexts. It provides detailed coverage of:

- OCR methods and their suitability for diverse German plate types.
- Operational example scenarios including exemptions, errors, and billing.
- Legal and privacy compliance measures aligned with GDPR and BDSG.
- System limitations and future work directions for scalability, robustness, and integration.

These foundations ensure a **robust technical, operational, and legal base** for subsequent methodology, implementation, testing, and evaluation chapters.